



RETVIN PANT

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EDUCATION

The University of Texas at Austin *B.S. Mechanical Engineering | Minor in Business*

Dec 2027

Relevant Coursework: Mechanics of Solids, Fluid Mechanics, Heat Transfer, Machine Elements, Mechatronics

Focus: Specialized Coursework and focus on Vehicle Design, Robotics Design and Aerospace Applications

GPA: 3.76

EXPERIENCE

Automation Engineering Intern – Daikin Comfort Technologies

May 2026 – August 2026

- Led a \$300K automation cell design project for heat exchanger manufacturing, defining cell layout, robot integration, material handling systems, and end-of-arm tooling with 53% production increase and \$305K+ in annual savings.
- Developed robot motion paths in FANUC software and designed material handling systems in SolidWorks to automate part transfer between forming and bending operations while reducing manual handling requirements.
- Implemented limit switch-based standby-mode for brazing equipment, reducing natural gas usage by ~10% per year.

Undergraduate Research Assistant - System Integration & Design Informatics Lab

August 2024 – Present

- Developing a custom robotic heated 3D printing bed utilizing SolidWorks and KiCAD, enhancing bed adhesion, and improving print arm odometry, streamlining manufacturing processes and reducing energy consumption by 5%.
- Devised a custom FDM extruder set up mountable to a 6 DOF arm using SolidWorks and Ansys, enabling conformal and inverted printing geometries not achievable with conventional fixed-head systems, reducing material waste by up to 85%.

Mechanical Engineering Intern - TEES Robotics and Automation Design Lab

May – August 2025

- Designed and manufactured a custom 2-foot shell mold utilizing SolidWorks and poka-yoke design techniques for RII, a spherical, inflatable rover for extraterrestrial purposes, improving assembly time in manufacturing settings by 45%.
- Built 10+ modular mounting solutions for RIII and RII iterations using SolidWorks.

ER4 Robotics Engineering Intern - NASA Johnson Space Center

June – August 2024

- Designed interchangeable wheel hubs for Microchariot in SolidWorks, enabling compatibility across 2-wheel variants.
- Engineered custom spring steel wheels for lunar mobility testing, validated via 5 mode FMEA for structural performance.

Mechanical and Software Engineering Intern - SPARX Engineering

August 2023 – June 2024

- Led the build of an automated candy sorter prototype with Onshape and Python to create a multistage self-sorting system.

LEADERSHIP AND ACTIVITIES

Co-Founder, Captain, Lead Dynamics Engineer - Longhorn Baja Racing

February 2025 – Present

- Founded the first Baja SAE Team at UT Austin, scaling the organization from 10 to 100+ active members across 8 subteams in 18 months, and securing faculty and industry support.
- Managed full-vehicle design CAD integration across 8 subsystems, conducting 10 formal design review cycles between suspension, drivetrain, chassis, and ergonomics prior to manufacturing release and ensuring SAE rule compliance.
- Designed custom double-wishbone suspension geometry in SolidWorks with 12 inches of wheel travel and -3° to $+1^{\circ}$ camber change and validated with ANSYS FEA for structural integrity under worst-case load conditions.
- Led development of the complete vehicle dynamics subsystem (suspension, steering, and brake systems) while directing 15+ engineers from concept through detailed design and owning systems-level tradeoff decisions and integration.

Vehicle Dynamics Engineer - Texas Guadalupe (Hyperloop Pod)

January 2025 – October 2025

- Applied Finite Element Analysis to optimize structural integrity under high-load cases, reducing overall weight by 5%.

Technical Team Captain; Design Lead - FRC Team 5414 – Pearadox

August 2021 – May 2024

- Directed all team technical aspects, overseeing an 80+ member team; led the robot arm and end effector system design.

SKILLS

CAD/CAE: SolidWorks, Onshape, Fusion 360, Ansys FEA, SolidWorks FEA, KiCAD, MasterCAM

Engineering Methods: GD&T, DFM, Tolerance Analysis, Jig and Fixture Design, FMEA, Poka-Yoke

Manufacturing: HAAS CNC Mill & Lathe, Sheet Metal, Rapid Prototyping, FDM & SLA 3D Printing, Shop Fabrication

Programming: Python, MATLAB, OpenCV, PyQt5, FANUC Roboguide Automation Software

Certifications: SolidWorks CSWA, HAAS Mill and Lathe, ACE Manufacturing